

Low Level Design Master Problem List

50
problems

50 must-solve problems · 6 categories · interview-ready

CORE OOP & REAL-WORLD SYSTEMS

7

- 01 Design Parking Lot
- 02 Design Vending Machine
- 03 Design ATM
- 04 Design Library Management System
- 05 Design Elevator System
- 06 Design Traffic Light System
- 07 Design Meeting Room Scheduler

PLATFORM & MARKETPLACE APPS

14

- 17 Design URL Shortener
- 18 Design BookMyShow
- 19 Design BookMyShow Seat Locking
- 20 Design Uber
- 21 Design Food Delivery Application
- 22 Design Online Hotel Booking System
- 23 Design Airline Management System
- 24 Design Restaurant Management System
- 25 Design Car Rental System
- 26 Design Amazon Order Management System
- 27 Design CricBuzz
- 28 Design Truecaller
- 29 Design Stock Exchange System
- 30 Design Learning Management System

DESIGN PATTERNS & ADVANCED LLD

10

- 41 Design Feature Flag System
- 42 Design Distributed ID Generator (Snowflake-like)
- 43 Design Circuit Breaker
- 44 Design Retry with Backoff Mechanism
- 45 Design Metrics and Monitoring System
- 46 Design Authentication System
- 47 Design Role-Based Access Control (RBAC)
- 48 Design Web Crawler (LLD focus)
- 49 Design Recommendation Engine (LLD focus)
- 50 Design Event-Driven Producer-Consumer System

GAMES & SIMULATIONS

3

- 08 Design Snake and Ladder Game
- 09 Design Tic-Tac-Toe Game
- 10 Design Chess Game

SOCIAL & COMMUNICATION APPS

6

- 11 Design Splitwise
- 12 Design Chat Application (WhatsApp-like)
- 13 Design Community Discussion Platform (Reddit-like)
- 14 Design LinkedIn (LLD focus)
- 15 Design Calendar Application
- 16 Design Online Voting System

INFRASTRUCTURE & CORE SYSTEMS

10

- 31 Design Cache System
- 32 Design Rate Limiter
- 33 Design Logging Framework
- 34 Design Notification System
- 35 Design Payment System (LLD)
- 36 Design File System
- 37 Design Task Scheduler
- 38 Design Search Autocomplete System
- 39 Design API Throttling System
- 40 Design Inventory Management System

★ ANCHOR PROBLEMS

Parking Lot · Elevator System · Chess Game · BookMyShow · Uber · Cache System · RBAC — [Unlock all 50 problems via shared patterns](#)